November 11, 2026

CURRICULUM VITAE

Conor Brown

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Dept. of Electrical & Computer Engineering
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EDUCATION

2025–Present **Ph.D. in Electrical & Computer Engineering, Vanderbilt University** (Expected 2030)
Advisor: Jack H. Noble. Dissertation research: Digital twin simulation of auditory perception with cochlear implants.

2024 **B.S. in Biomedical Engineering, University of Tennessee, Knoxville**
GPA: 4.0 / 4.0. Graduated Summa Cum Laude. Top Graduate Award (2024) and Outstanding Senior Award (2023).

RESEARCH EXPERIENCE

2025–Present **Graduate Research Assistant, Noble Lab, Vanderbilt University**

- Develop differentiable computational models of the auditory pathway and learned, patient-specific stimulation strategies to quantify and improve how faithfully cochlear implants convey sound.
- Rewrote the laboratory's auditory-nerve model in PyTorch, verified it against the original implementation, and made it differentiable, so that stimulation can be optimized through each patient's digital twin.
- Built an end-to-end differentiable encoder in PyTorch that maps audio directly to electrode selections and pulse currents, optimized per patient and exportable as a hardware-realizable pulse table.
- Designed convolutional decoders that reconstruct speech from the simulated neural firing of patient-specific digital twins. Decoders that never trained on a patient's twin reconstruct that patient's speech nearly as well as decoders trained on it (8 twins), the basis of a per-patient measure of speech transmission (SPIE Medical Imaging 2027).
- Trained decoders with no target sound, using only whether an automatic speech recognizer understood their output; an independent recognizer's word identification rose from 24% to 73% across the 8 twins.
- Developed Transformer models predicting STFT phase from log-magnitude spectra, evaluated with MFCC-based metrics.

2023–2025 **Research Assistant, SMART Lab (PI: Prof. Xiaopeng Zhao), University of Tennessee, Knoxville**

- Built AI-driven digital-health tools for dementia care — a RAG chatbot (LangChain, OpenAI, Pinecone), assistive web applications, and social-robotics control systems — collaborating across interdisciplinary engineering and research teams.

FELLOWSHIPS AND TRAINING SUPPORT

2026–Present **Predoctoral Trainee, NIH/NIBIB T32 Training Program for Innovative Engineering Research in Surgery and Intervention (T32EB021937), Vanderbilt University**
Ruth L. Kirschstein National Research Service Award institutional training grant. PD/PI: Michael I. Miga.

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CLINICAL EXPERIENCE

2022 **Medical Assistant, Camai Community Health Center, Naknek, AK**

- Performed phlebotomy, IV medication administration, wound debridement, radiography, suturing, and laboratory testing in primary and emergency care; served as the clinic's go-to resource for troubleshooting medical technology.

HONORS AND AWARDS

2026 Best Teaching Assistant Award, Dept. of Electrical & Computer Engineering, Vanderbilt University

2024 REACH project selected as a UT Grand Challenges Grant awardee (Type 1, up to $100,000) — Undergraduate Research Assistant

2024 Top Graduate Award, Dept. of Biomedical Engineering, University of Tennessee, Knoxville

2024 Graduated Summa Cum Laude, University of Tennessee, Knoxville

2024 B.S. conferred with a 4.0 / 4.0 cumulative GPA

2023 Outstanding Senior Award, Dept. of Biomedical Engineering, University of Tennessee, Knoxville

TEACHING EXPERIENCE

2025–2026 **Teaching Assistant, Dept. of Electrical & Computer Engineering, Vanderbilt University**
ECE 4951 – Electrical and Computer Engineering Design (Spring 2026)
ECE 2218 – Microcontrollers (Fall 2025)
Recognized with the 2026 ECE Best Teaching Assistant Award.

2022–2023 **Tutor, Academic Success Center, University of Tennessee, Knoxville**
Mentored students in engineering, programming, organic chemistry, and calculus.

PUBLICATIONS

Peer-Reviewed Journal Publications

[1] Tyler Morris, **Conor Brown**, Xiaopeng Zhao, Linda Nichols, Jennifer Martindale-Adams, Sharon Bowland, Wenjun Zhou, "Transforming dementia caregiver support with AI-powered social robotics," *Frontiers in Robotics and AI*, vol. 12, 1704313, 2026. DOI: 10.3389/frobt.2025.1704313.

Peer-Reviewed Conference Proceedings

[2] **Conor J. Brown**, Jack H. Noble, "Decoding simulated auditory nerve activity to measure cochlear implant speech transmission from patient-specific neural health," *Proceedings of SPIE Medical Imaging 2027: Image-Guided and Robotic Interventions*, paper 14520-75, accepted for poster presentation, Vancouver, BC, February 2027.

[3] Tyler Morris, **Conor Brown**, Seungwoo An, Jeremiah Augustine, Andrew Ward, Laura Enomoto, Erin Campbell, Xiaopeng Zhao, "Using Social Robots and AI to Perform Genetic Risk Assessment for Cancer," *Proceedings of the AAAI Symposium Series*, vol. 4(1), pp. 359–366, 2024. DOI: 10.1609/aaaiss.v4i1.31817.

Research Contributions

- Designed and built the social-robot control system used in:

  Fengpei Yuan, Robert Bray, Michael Oliver, Joshua Duzan, Monica Crane, Xiaopeng Zhao, "A Social Robot-facilitated Performance Assessment of Self-care Skills for People with Alzheimer's: A Preliminary Study," *International Journal of Social Robotics*, vol. 16(9), pp. 2065–2078, 2024. DOI: 10.1007/s12369-024-01174-6.

- Performed cell counting, tissue culture maintenance, and trichrome staining for two myocardial tissue-engineering studies in the laboratory of Prof. Jianyi Zhang (University of Alabama at Birmingham):

  Pretorius D, Kahn-Krell AM, LaBarge WC, Lou X, Kannappan R, Pollard AE, Fast VG, Berry JL, Eberhardt AW, Zhang J, "Fabrication and characterization of a thick, viable bi-layered stem cell-derived surrogate for future myocardial tissue regeneration," *Biomedical Materials*, vol. 16(3), 2021. DOI: 10.1088/1748-605X/abc107.

  Pretorius D, Kahn-Krell AM, LaBarge WC, Lou X, Zhang J, "Engineering of thick human functional myocardium via static stretching and electrical stimulation," *iScience*, vol. 25(3), 103824, 2022. DOI: 10.1016/j.isci.2022.103824.

CONFERENCE PRESENTATIONS AND POSTERS

[1] **Conor J. Brown** and Jack H. Noble, "Transformer Model to Predict Delta Phase from Log Magnitude STFT for CI Processing," poster presented at the VISE Annual Surgery, Intervention, and Engineering Symposium, Nashville, TN, 2025.

[2] **Conor J. Brown** and Jack H. Noble, "Transformer Model to Predict Delta Phase from Log Magnitude STFT for CI Processing," poster presented at Vanderbilt ECE Day, Nashville, TN, 2026.

[3] **Conor J. Brown**, "Attention-Focused Cochlear Implants," technology pitch presented at the VISE Technology Pitch, Vanderbilt Institute for Surgery and Engineering, Nashville, TN, 2026.

[4] **Conor J. Brown** and Jack H. Noble, "Decoding simulated auditory nerve activity to measure cochlear implant speech transmission from patient-specific neural health," poster accepted for presentation at SPIE Medical Imaging 2027, Vancouver, BC, February 2027.

[5] **Conor J. Brown** and Tyler Morris (PI: Xiaopeng Zhao), technology demonstration and presentation of the REACH (Rural Expansion of Access to Genetic Counseling and Healthcare through AI Innovations) project, UT Grand Challenges Summit, Murfreesboro, TN, February 2025.